NACS 645 – Why do we cooperate?

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Valentin Guigon

DEPARTMENT OF PSYCHOLOGY

SLD Lab

PROGRAM IN NEUROSCIENCE & COGNITIVE SCIENCE

Cooperation, competition, coordination

Prisoners vs civilians, rigged boats, *The Dark Knight*

Mexican standoff, *Reservoir dogs*

	C	*D*
C	-1,-1	-1,0
D	0,-1	-1,-1

	C	*D*
C	1,1	-1,0
D	0,-1	-1,-1

Cooperation, competition, coordination

Prisoners’ dilemma

		C	*D*
P1	*C*	-1,-1	-4,0
	D	0,-4	-3,-3

No matter what P2 plays, it is best to defect (1 Nash Equilibrium)

Mixed motive: cooperation & competition

Side of the road

If P2 goes to left, P1 better go left; same for the right (2 NE)

	Left	*Right*
Left	1,1	0,0
Right	0,0	1,1

Pure coordination: exactly aligned interests

Battle of the sexes

		B	*F*
P1	*B*	2,1	0,0
	F	0,0	1,2

The best response is to select the action that the other wants (2 NE)

Mixed motive: cooperation & competition

Matching pennies

P1 ends up in a circle (0 NE)

	Heads	*Tails*
Heads	1,-1	-1,1
Tails	-1,1	1,-1

Pure competition: exactly opposed interests

Cooperation, competition, coordination

Prisoners' dilemma

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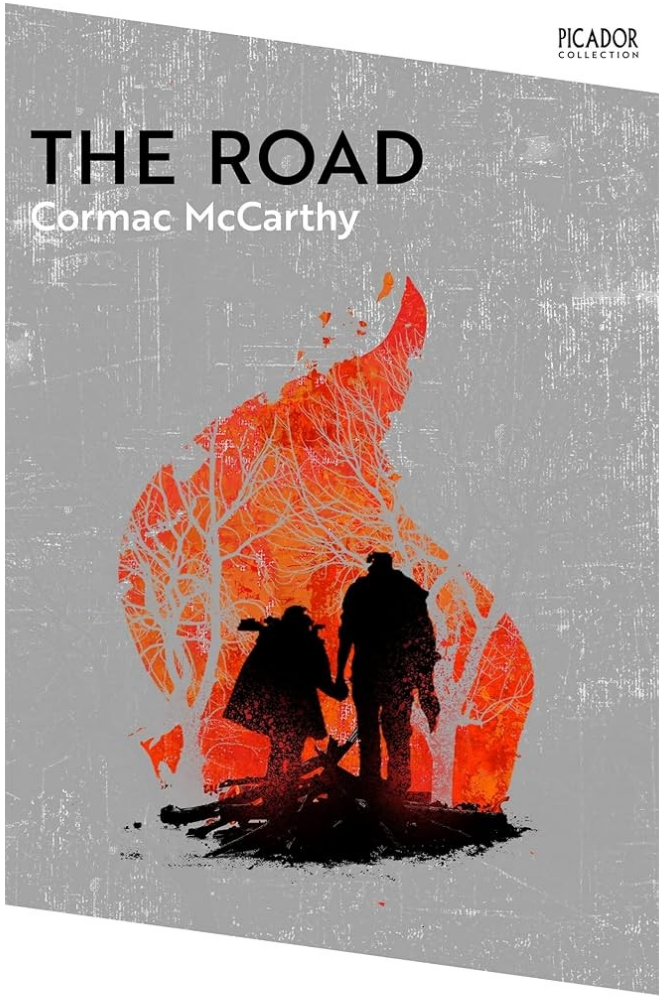
Mixed motive: cooperation & competition

- Cooperation happens in situations with conflicting interests of actors: actors opt for an action suboptimal for themselves but superior for the collective.
- The population does best if individuals cooperate (*higher social welfare*), but for each individual there is a temptation to defect (*higher individual utility*).
- One individual pays a cost for another to receive a benefit.
- Problem in the PD: why should you reduce your own fitness to increase that of a competitor in the struggle for survival?

Choosing actions

- What will other players do?
- What should I do in response?



Classical rationality

Homo economicus

- Rational agents
- Perfectly informed
- Maximize their utility

Rational choice theory

- Coherent (ordered) preferences
- Rational thinking leads to choices aligned with preferences
- Group behavior reflect the aggregate of individual behaviors (efficient allocation)

Mill, Pareto, Samuelson, etc.

Game theory

Assumptions:

- Each agent has its own description of states of the world
- Each agent has a utility function (preferences, uncertainty profile)
- Each agent maximizes expected utility (decision-theoretic rationality)

Ingredients to describe decision-making:

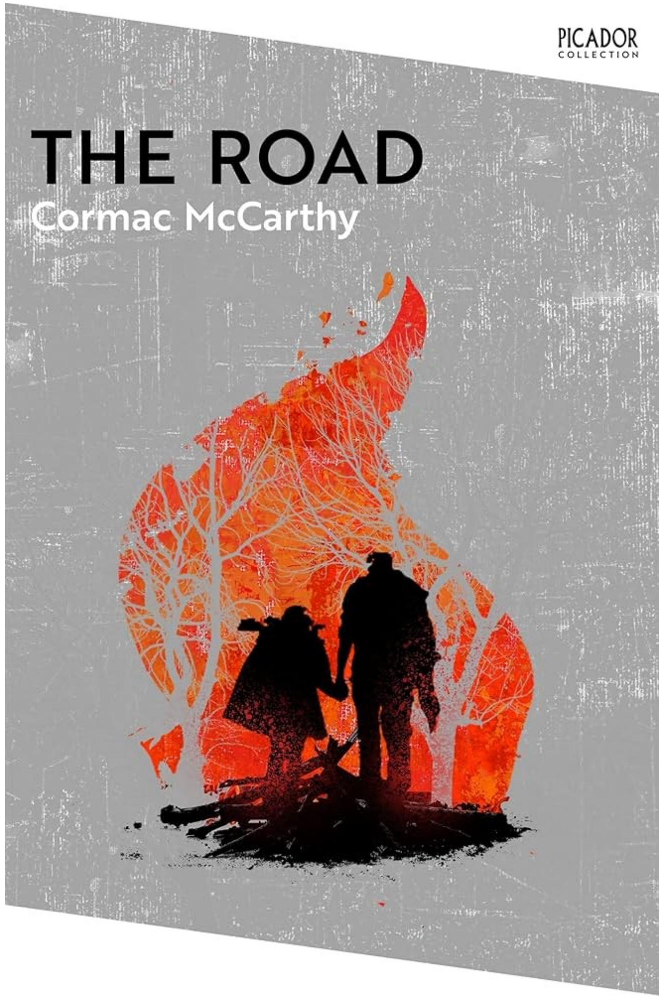
- Set of Players (people, governments, companies)
- Set of Actions (bid, strike, vote, cooperate, defect)
- Set of Payoffs (monetary preferences, social preferences)

Choosing actions

- What will other players do?
- What should I do in response?

Given prior assumptions (each agent has its own description of states of the world, a utility function and aims at maximizing expected utility):

Each player ***best responds*** to the others (*Nash equilibrium*)



Nash equilibrium

Side of the road

	Left	*Right*
Left	1,1	0,0
Right	0,0	1,1

An equilibrium:

- A list of actions (action set)
- With which each player's action maximizes his/her payoff given the actions of the others
- In a consistent/stable pattern (profile)

Implications:

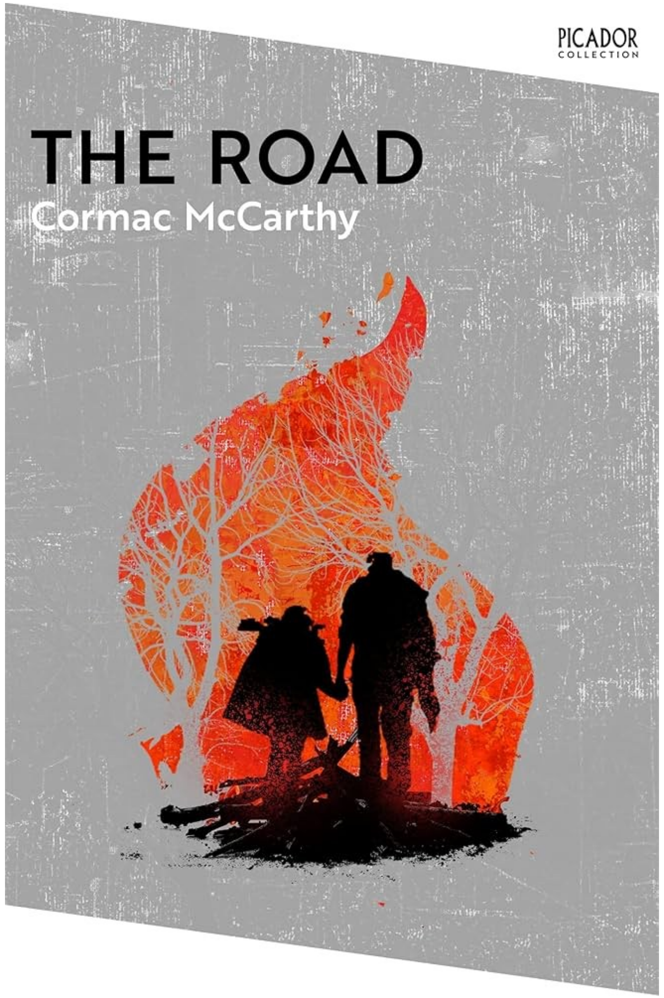
- Nobody has an incentive to *deviate* from their action if an equilibrium profile is played
- Someone has an incentive to *deviate* from a profile of actions that do *not* form an equilibrium

Under correct assumptions, equilibria should be expected to be played once participants understand the game (non-equilibria should vanish over time)

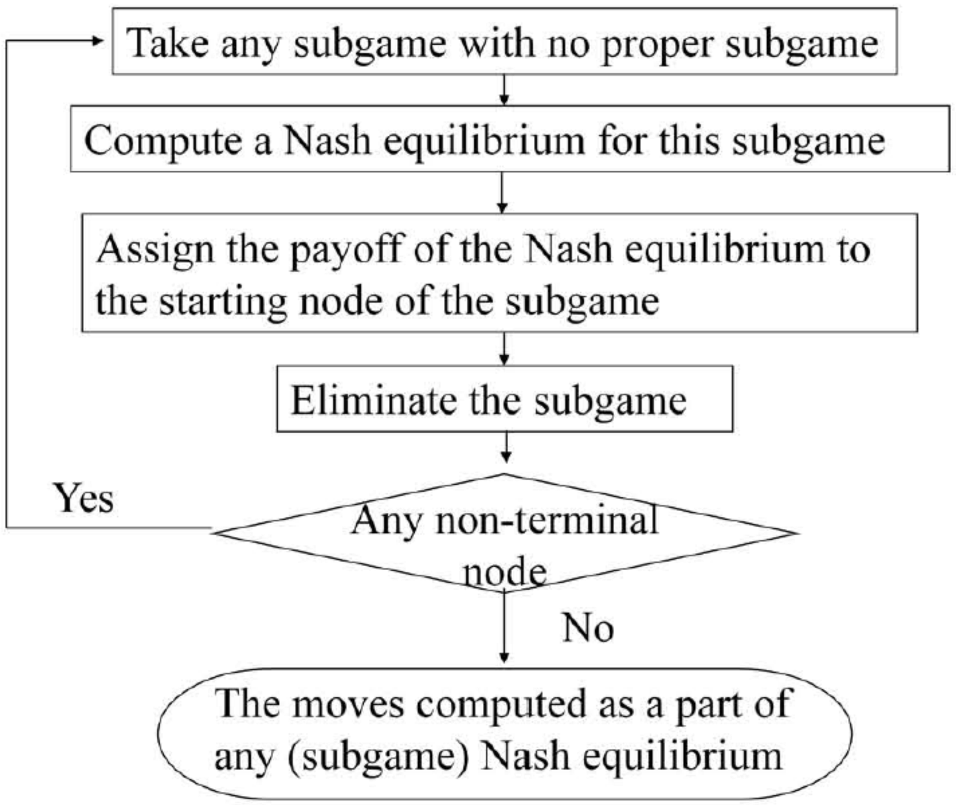
Choosing the best action

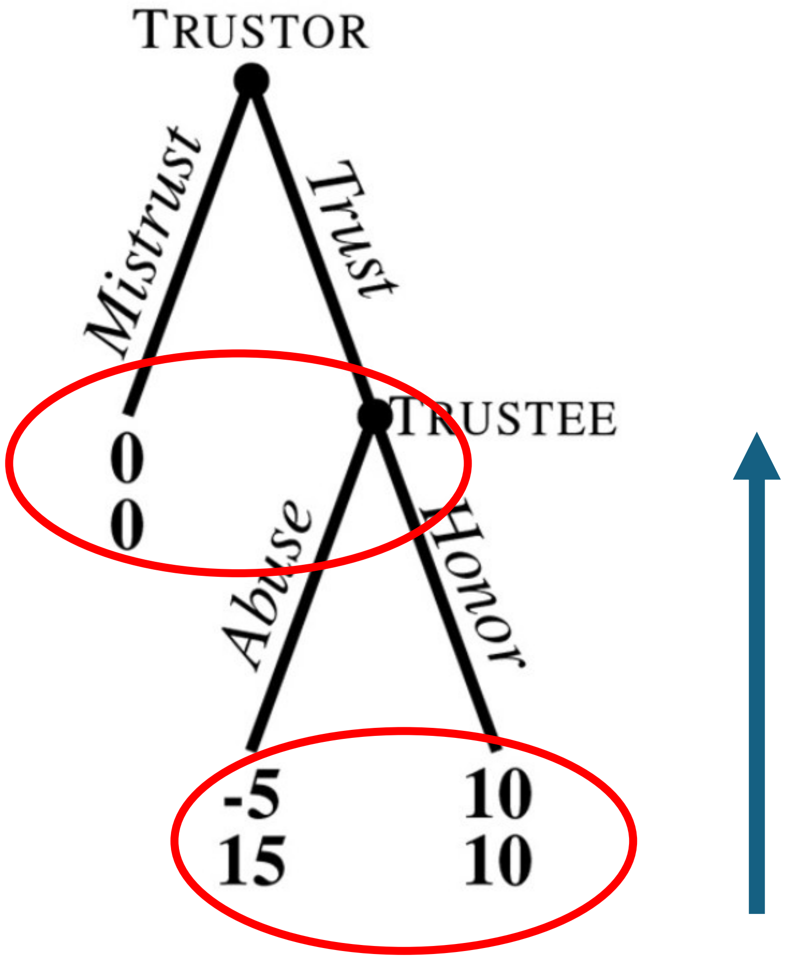
- If we knew what everyone else was going to do, it would be easy to pick the action
- Idea: look for stable action profiles (actions that nobody has incentives to deviate from)
- The « pure strategy » Nash equilibrium is a set of actions, one for each agent, such that each action is the best response to the actions of others

- In *The road*, the world has ended and cannibalism is widespread. We can assume the best action is the *untrustworthy* one
- In the *State of nature*, we may need others and we want to avoid risks. We can assume the best action is the *trustworthy* one



Choosing actions – backward induction

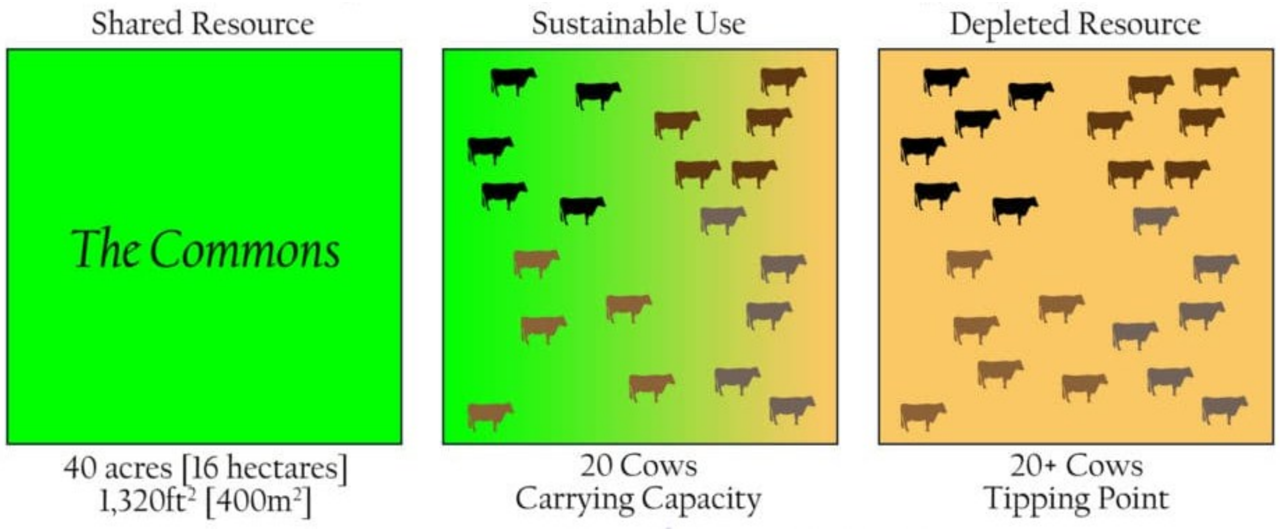




If we can prevent the defection at the last trial n (and all n_{-i} trials), we can incentivize cooperation (e.g., infinitely and indefinitely repetead games)

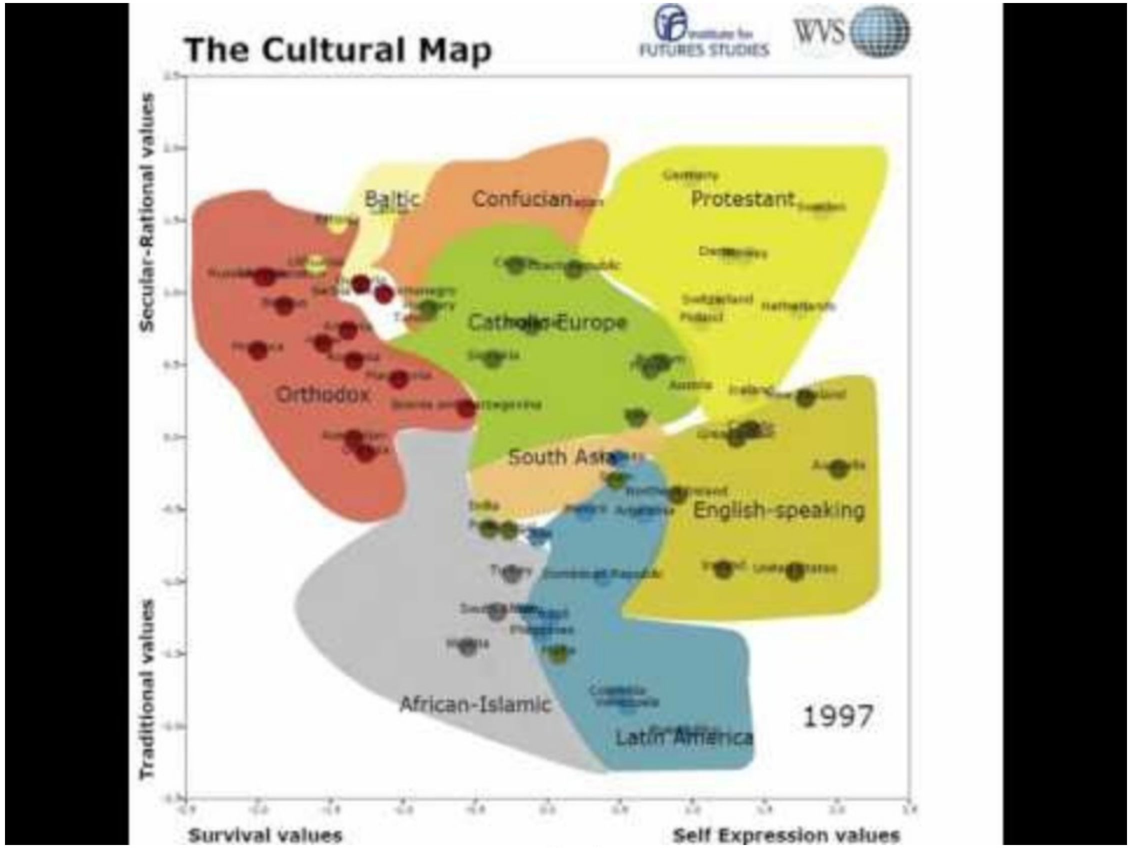
Cooperation at the population-level

Joshua Greene, Moral Tribes (2013). *Penguin Press*



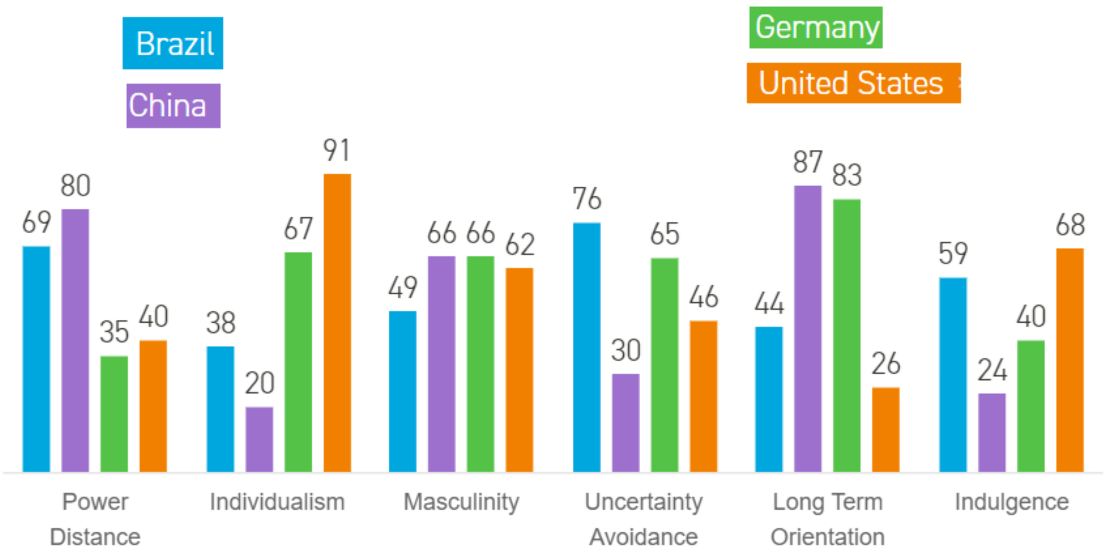
- Hardin: We need *mutual coercion mutually agreed upon*
- Greene: *Human morality evolved* to align individual self-interest with collective welfare
    - Shared norms, emotions, reputations make selfish behavior costly. Guilt, shame, pride, and gratitude act as internal “moral technologies”
- Rand, Nowak: Evolution gives rise to people who are truly altruistic and cooperate (*Intuitive reciprocation*)

Inglehart-Welzel culture map



1. *Traditional vs secular–rational values*: traditional values emphasize religion, parent–child ties, deference to authority, absolute standards; secular-rational values emphasize more acceptance of divorce, abortion, euthanasia, etc.
2. *Survival vs self-expression values:* survival values emphasize economic/physical security, ethnocentrism, low trust/tolerance; self-expression values emphasize subjectively defined well-being, tolerance of diversity, participatory decision-making, environmental protection.

Hofstede's cultural dimensions theory



- **Power distance**: describes the extent to which less powerful members of a society accept and expect unequal distribution of power.
- **Individualism vs collectivism**: measures whether people in a culture prioritise personal goals over group goals, or vice versa.
- **Masculinity vs femininity**: Masculinity refers to cultures that value competitiveness, achievement, and material success, while femininity values cooperation, care, and quality of life.
- **Uncertainty avoidance**: measures how comfortable a culture is with ambiguity, change, and the unknown.
- **Long-term avoidance**: reflects whether a culture prioritises future rewards over immediate results.
- **Indulgence vs restraint**: looks at how freely societies allow people to gratify their desires and enjoy life.

Can we predict that we will observe cooperation
to the relatively same extent,
in distinct societies,
and despite their distinction in values,
if we assume the same set of mechanisms for the evolution of cooperation?

In other words, does the set of 5 mechanisms for the evolution of cooperation guarantee cooperation everywhere?